

CSE4502 (Operating Systems Lab 9): 1B

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Instruction: Deadlock Detection in Single-Instance Resource Systems

You will implement an algorithm that analyzes a Resource Allocation Graph (RAG), and check if the system exhibits deadlock.

1 Resource Allocation Graph (RAG)

A RAG is a directed graph used to represent the state of the system:

- Nodes: Two types of nodes exists: Processes ($P = \{P_1, P_2, \dots\}$) and Resources ($R = \{R_1, R_2, \dots\}$).
- Edges:
 - Request Edge ($P_i \rightarrow R_j$): Process P_i wants resource R_j .
 - Assignment Edge ($R_j \rightarrow P_i$): Resource R_j is currently held by process P_i .

2 Input Specification

Your program will read from standard input (or a file) in the following format:

1. Line 1: Two integers N (number of processes) and M (number of resources).
 - Processes are named 0 to N-1.
 - Resources are named 0 to M-1.
2. Line 2: An integer E representing the total number of edges (requests + assignments)
3. Next E lines have three values, $u \ v \ type$
 - u : Source node ID.
 - v : Destination Node ID
 - $type$: A string, either "R" (Process u requests Resource v) or "A" (Resource u is assigned to Process v).

2.1 Example Input

```
1 3 3
2 5
3 0 0 R
4 0 1 A
5 1 1 R
6 1 2 A
7 2 2 R
```

2.2 Interpretation

- P0 requests R0.
- R0 is assigned to P1.
- P1 requests R1.
- R1 is assigned to P2.
- P2 requests R2.

2.3 Output

```
1 NO DEADLOCK
```